SeGlo-12: A Multimodal Glove Integrating Flex Sensors and Computer Vision for Sign-to-Speech Translation

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Abstract. This study presents SeGlo-12, a dual-mode sign language translation system designed to bridge communication barriers between deaf or hard-of-hearing individuals and hearing individuals unfamiliar with sign language. The system integrates both sensor-based and vision-based technologies to recognize and translate American Sign Language (ASL) gestures into text and synthesized speech in real time. SeGlo-12 features two independent input modes: a smart glove equipped with flex sensors and an MPU-6050 gyroscope for rule-based static gesture recognition, and a smartphone-based vision module using MediaPipe Hands and a TensorFlow Lite classifier for machine learning–based gesture recognition. Both modes are integrated into an offline-capable Android mobile application that provides real-time multimodal output through text display and speech synthesis. Evaluation results showed that the vision-based module achieved an overall accuracy of 87% and a macro-averaged F1-score of 0.86 across 38 gesture classes. Beta testing with ten participants yielded an overall user satisfaction score of 4.51/5, with positive feedback regarding usability and accessibility. The findings demonstrate that combining wearable sensing and lightweight computer vision techniques can provide a practical and accessible assistive communication solution. Future improvements include dynamic gesture recognition, adaptive sensor classification, and multimodal fusion for enhanced robustness and scalability.

Keywords: Sign language recognition · Wearable technology · Assistive communication · American Sign Language · Real-time translation

1 Introduction

Communication is a fundamental human need, yet millions of people who are deaf or hard of hearing face daily barriers when interacting with the wider hearing population. According to the World Health Organization, more than 360 million people worldwide live with disabling hearing loss [1]. For many of them, sign language is the primary mode of communication. However, the majority of hearing individuals do not understand sign language, leading to frequent misunderstandings, limited opportunities, and feelings of exclusion in both social and professional contexts. This persistent

communication gap highlights the urgent need for inclusive technologies that can bridge the divide between signers and non-signers.

To address this challenge, researchers have explored sign language recognition (SLR) systems that automatically translate gestures into text or speech. Two primary approaches dominate the field. **Sensor-based systems**, often implemented with wearable gloves, use flex sensors and inertial units to capture finger movements and hand orientation. These systems are compact and resistant to environmental noise but typically recognize only static gestures and can be uncomfortable or costly to wear for extended periods. In contrast, **vision-based systems** use cameras and machine learning to interpret gestures from video input. Advances such as MediaPipe and deep learning have enabled accurate, real-time recognition of dynamic gestures, but vision-only systems remain sensitive to factors like lighting conditions, background clutter, and line-of-sight restrictions. Although both approaches show promise, each has limitations that reduce their effectiveness in real-world use.

Although the challenge of sign language recognition has seen substantial advances, the vast majority of systems still utilize a single modality—either glove-based or vision-based—and this has left systems with intrinsic drawbacks of both approaches. While hybrid approaches do exist, they are in the minority and are limited to prototypes, essentially proof-of-concept systems rather than practical, deployable systems. Furthermore, there are very few efforts pointing toward mobile-first, and offline-capable designs that utilize mobile accessibility options beyond the cloud (given that cloud resources require internet connectivity). This continues to create a gap between research advances and the real-world needs of deaf and hard of hearing communities.

To bridge this gap, this paper presents **SeGlo-12 (Sensor Gloves)**, an innovative dual-mode translator that integrates both sensor-based and vision-based recognition into a single mobile application. The system combines an Arduino-powered glove, equipped with flex sensors and a gyroscope for capturing static signs, with a smartphone-based computer vision pipeline that uses MediaPipe and lightweight neural networks to detect static gestures. By fusing these two modalities, SeGlo leverages the strengths of each while mitigating their weaknesses, ensuring more robust and flexible communication support. Unlike prior works that focus narrowly on either gloves or cameras, SeGlo emphasizes **practicality, offline operation, and user-centered design**, making it more suitable for real-world adoption. Initial user testing further suggests strong potential for SeGlo to function as an accessible assistive tool that empowers deaf and hard-of-hearing individuals in daily communication.

2 Related Works

SLR systems generally fall into three categories: sensor-based systems, vision-based systems, and hybrid systems. Sensor-based approaches commonly utilize wearable gloves equipped with flex sensors and inertial measurement units (IMUs), while vision-based systems employ computer vision and machine learning techniques to recognize gestures from camera input. Hybrid systems attempt to combine the strengths

of both modalities to improve robustness, accessibility, and usability in real-world environments.

Sensor-based SLR systems have demonstrated strong robustness against environmental factors such as lighting and background interference. Ahmed et al. [2] reviewed glove-based recognition systems and identified flex sensors and IMUs as commonly used hardware for capturing static hand gestures. Recent implementations by Dalal et al. [6], Hammad and Fadel [7], and Mishra et al. [9] further validated the effectiveness of combining flex sensors, accelerometers, and Arduino-based platforms for real-time gesture translation. These systems demonstrated reliable recognition under varying environmental conditions and confirmed the feasibility of low-cost wearable assistive devices. However, most glove-based systems remain limited by hardware dependence, user comfort concerns, and restricted support for dynamic gestures. In addition, many existing implementations rely on rule-based classification strategies with limited scalability and adaptability to inter-signer variation.

Vision-based approaches have increasingly adopted deep learning and computer vision frameworks to improve gesture recognition accuracy and real-time performance. Al-Mohimeed et al. [3] utilized ConvLSTM architectures for dynamic gesture recognition, while Alsharif et al. [4] combined YOLO and MediaPipe keypoint tracking to achieve highly accurate real-time American Sign Language interpretation. Similarly, Singh et al. [11] demonstrated the effectiveness of MediaPipe landmark extraction combined with convolutional neural networks for ASL classification. Ciraolo et al. [5] further showed that lightweight landmark-based recognition pipelines can operate efficiently on low-resource devices. Despite these advances, vision-based systems remain sensitive to lighting conditions, occlusion, and background complexity, which may reduce reliability in uncontrolled environments.

Recent studies have highlighted the potential of hybrid and multimodal approaches for improving sign language recognition systems. Manasa et al. [8] demonstrated that combining MediaPipe landmark extraction with LSTM architectures enables more effective temporal gesture recognition, while Najib [10] emphasized the importance of integrated systems that combine gesture recognition, facial expressions, and mobile accessibility features. However, most existing studies remain limited to proof-of-concept prototypes or single-modality systems. Few works focus on offline-capable, mobile-first implementations that integrate both glove-based and vision-based recognition into a unified application. To address this gap, SeGlo-12 combines sensor-based and vision-based recognition in a dual-mode mobile platform designed for accessibility, portability, and real-time offline translation.

3 Materials and Methods

The SeGlo system was designed as a dual-mode sign language translator, integrating both **sensor-based glove recognition** and **vision-based camera recognition** into a

single mobile application. The methods followed in this project encompass hardware design, software pipeline implementation, mobile integration, and error-handling strategies.

3.1 System Overview

SeGlo operates in two modes: **Glove Mode** and **Camera Mode**. In Glove Mode, an Arduino Nano-based smart glove equipped with flex sensors and an inertial measurement unit (IMU) captures static hand gestures and transmits data via Bluetooth (HC-05) to the smartphone application. In Camera Mode, the mobile device uses MediaPipe to extract 21 3D hand landmarks per frame, which are then classified by a TensorFlow Lite model. Both modes output recognized gestures as **spoken audio** and **text display**, ensuring accessibility in real time. The block diagram of the system is illustrated in Fig. 1.

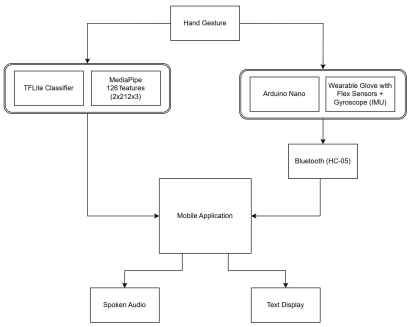


Fig. 1. Overall system block diagram of the SeGlo-12 platform

3.2 Hardware Design (Smart Glove Module)

The Smart Glove is constructed on an Arduino Nano microcontroller and integrates five flex sensors, one per finger, to measure bending angles, along with a gyroscope/accelerometer (MPU-6050) for capturing orientation and hand motion. Sensor signals are read via the Arduino's analog inputs, preprocessed (normalization and calibration), and mapped to predefined gesture labels using hardcoded threshold logic. The recognized gesture label is transmitted via an HC-05 Bluetooth module to the mobile application for further processing. This approach ensures robustness against environmental interference such as lighting and background noise, making the glove reliable for static ASL letter detection. Fig. 2 illustrates the final prototype.

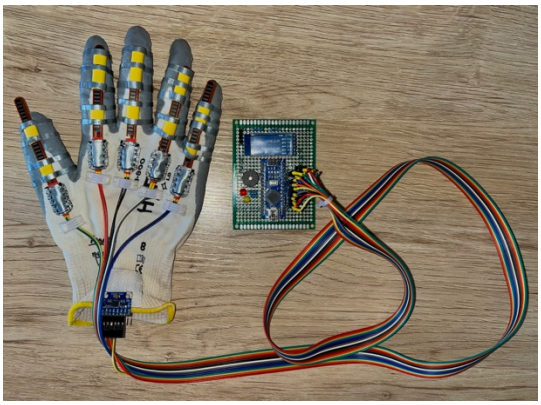


Fig. 2. Smart glove prototype with flex sensors and MPU-6050 module

3.3 Camera-Based Recognition

For camera-based recognition, the system employs MediaPipe Hands to extract 21 keypoints per hand, generating a 126-dimensional feature vector (2 hands × 21 landmarks × 3 coordinates). These features are input into a TensorFlow Lite classifier consisting of a Dense Neural Network (DNN), optimized for real-time inference on mobile devices. This pipeline enables recognition of static ASL gestures. While dynamic gestures remain a challenge, the modular design allows for future integration of temporal models such as LSTM networks.

3.4 Mobile Application

The Android mobile application (developed using Kotlin and Jetpack Compose) integrates both glove and camera recognition pipelines. It manages Bluetooth communication with the glove, executes the TensorFlow Lite inference pipeline, and provides multimodal outputs in real time. Recognized signs are displayed as text on screen and simultaneously vocalized through Android's Text-to-Speech (TTS) API. Users may switch between Glove Mode and Camera Mode through the app interface, depending on preference and environment. The application also includes error-handling mechanisms such as Bluetooth reconnection prompts and no-hand-detected alerts for improved usability. Fig. 3 illustrates the user interface of the mobile application.

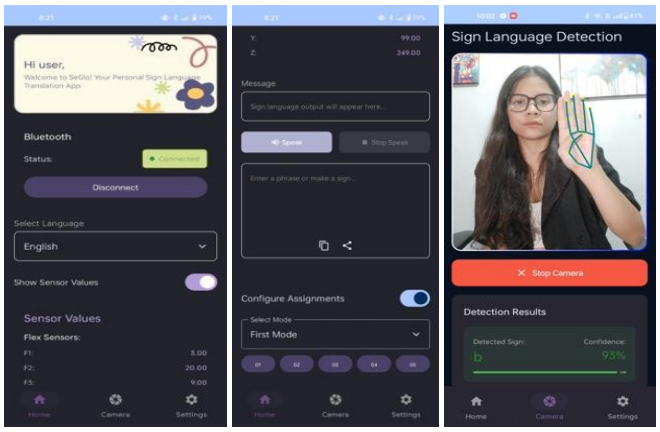


Fig. 3. Android mobile application interface for glove and camera modes

3.5 System Flow and Architecture

The functional flow of SeGlo is presented in Fig. 4 (Flowchart). In Glove Mode, sensor data is calibrated, mapped, and transmitted to the app via Bluetooth. In Camera Mode, MediaPipe extracts landmarks which are classified by the TFLite model. Both modes converge in the Decision Manager, which applies confidence thresholds before outputting text or speech. Error handling mechanisms (e.g., reconnect prompts, retry loops) ensure robustness. Fig. 5 (System Architecture) details the layered architecture, including input capture, preprocessing, inference, communication, and output modules.

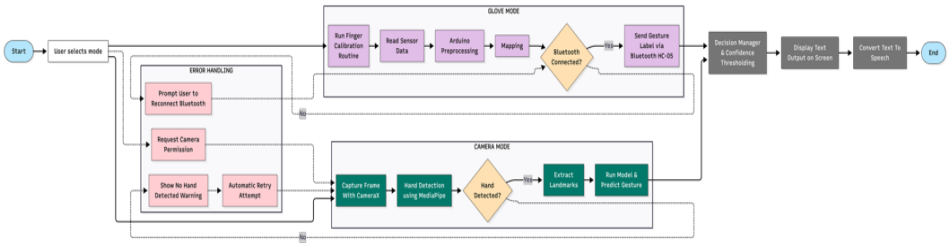


Fig. 4. Operational flowchart of the SeGlo translation system

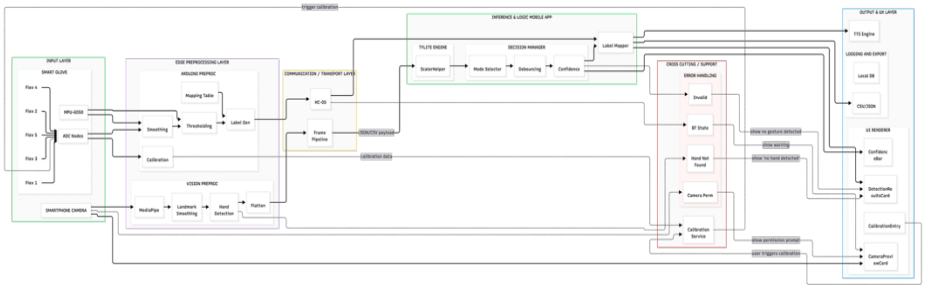


Fig. 5. Layered system architecture for multimodal gesture recognition

3.6 Data Processing and Gesture Classification

In Glove Mode, sensor readings obtained from the flex sensors and MPU-6050 inertial measurement unit are mapped to predefined gestures using a hardcoded rule-based classification strategy. This lightweight implementation enables real-time processing while maintaining low computational overhead on the Arduino Nano platform.

In Camera Mode, gesture recognition is performed using a TensorFlow Lite classifier trained on MediaPipe hand landmark data. The model is optimized for efficient inference on mobile devices while preserving recognition accuracy for static ASL gestures.

To reduce false positives and misclassifications, a confidence thresholding mechanism is applied before recognized gestures are displayed or converted into speech output.

4 Results and Discussion

The SeGlo system was evaluated through two complementary approaches: machine learning model evaluation and beta testing. For the vision-based module, a held-out test set of 1,950 samples spanning 38 gesture classes was independently collected using the same MediaPipe hand-tracking pipeline employed during training. Raw landmark vectors (126 floats: 2 hands × 21 landmarks × 3 axes) were extracted, scaled using the original training StandardScaler, and passed through the trained neural network for classification. Four standard metrics were computed per class: Precision, Recall, F1-Score, and Support. For user evaluation, ten participants were recruited, including one member of the hard-of-hearing community—to assess the prototype through structured beta testing. The system was tested on a wearable glove prototype equipped with five flex sensors, an MPU-6050 gyroscope/accelerometer, and an Arduino Nano, connected via HC-05 Bluetooth to an Android smartphone application. Metrics collected included recognition accuracy, latency, usability, and user satisfaction ratings. Four standard metrics were computed per class. Precision measures the proportion of correct positive predictions (Eq. 1), Recall captures the proportion of actual positives correctly identified (Eq. 2), and F1-Score balances the two as their harmonic mean (Eq. 3).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

4.1 Vision-Based Model Performance

The vision-based gesture recognition module achieved an overall accuracy of 87% across 38 gesture classes on the held-out test set. The macro-averaged F1-score of 0.86 and weighted F1-score of 0.87 are nearly identical, confirming that strong performance is maintained across both high-frequency ASL letters and less frequent conversational phrases. This balance is significant for a real-world assistive system where no gesture class should be systematically neglected. Table 1 summarizes the aggregate performance, while the full class-wise breakdown is visualized in Fig. 6.

Table 1. Aggregate vision-based model performance (38 classes, 1,950 test samples)

Averaging Method	Precision	Recall	F1-Score	Total Samples
Macro Average	0.90	0.85	0.86	1,950
Weighted Average	0.91	0.87	0.87	1,950
Overall Accuracy		**0.87**		1,950

Seven gestures—c, f, n, p, q, s, and w—achieved a perfect F1-score of 1.00, indicating zero misclassifications across the entire test set. An additional eleven gestures, including the conversational signs which (0.97), how (0.97), and more (0.91), scored above F1 = 0.90. This demonstrates that the model generalizes effectively beyond the ASL alphabet to context-bearing phrases, which is a key contribution of this work. The class-wise metrics bar chart (Fig. 6) provides a visual overview of this distribution, clearly showing the performance tiers across all 38 gestures.

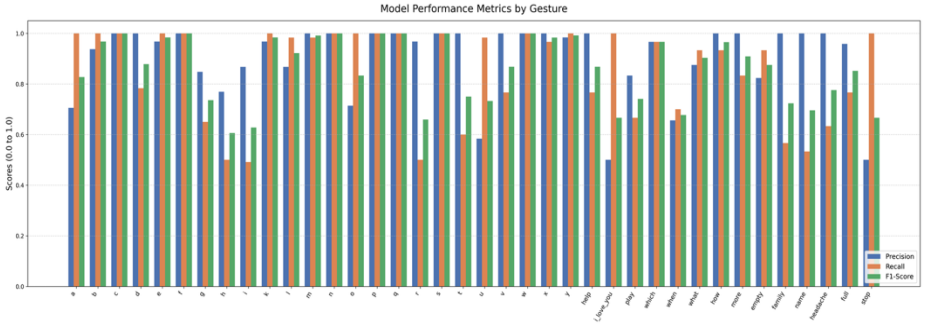


Fig. 6. Class-wise precision, recall, and F1-score across 38 ASL gesture classes

4.2 Confusion Matrix Analysis

The confusion matrix (Fig. 7) reveals the spatial distribution of classification errors and corroborates the quantitative metrics. The dominant diagonal confirms strong overall recognition, while the off-diagonal concentrations highlight the key failure modes. The most prominent error cluster involves the gesture pair i and i_love_you. Both signs share an extended pinky finger as a sub-component, creating high spatial landmark overlap in the MediaPipe feature vector. This confusion is asymmetric: i_love_you achieves a Recall of 1.00 but a Precision of only 0.50, meaning the model captures every true i_love_you occurrence but at the cost of falsely predicting i_love_you for many actual i samples. The gesture i consequently suffers a Recall of only 0.49 and an F1-score of 0.63, making it the most challenging class in the dataset. Similarly, h and r both exhibit recall collapses to 0.50. Both involve two adjacent extended fingers with close inter-finger proximity, making their 3D landmark representations highly sensitive to minor wrist rotation or signer-specific variation. The high precision for both (0.77 and 0.97 respectively) confirms that when the model does predict these gestures, it is almost always correct—but it systematically underdetects them. These patterns are consistent with documented challenges in fine-grained hand gesture discrimination in the computer vision literature [4, 11].

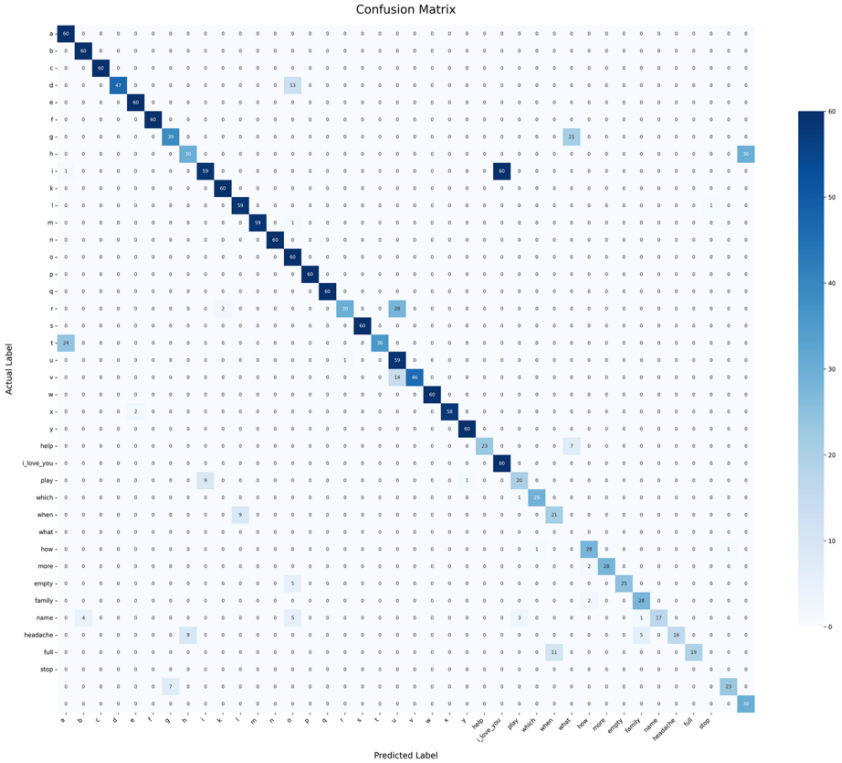


Fig. 7. Confusion matrix across 38 ASL gesture classes. Darker diagonal cells indicate higher correct classification rates.

4.3 Multi-Class ROC Curve Analysis

The multi-class ROC curves (Fig. 8) complement the per-class metrics by illustrating the trade-off between true positive and false positive rates across all gesture classes. The macro-averaged AUC and the tight clustering of individual class curves near the upper-left corner indicate strong discriminative capacity across the gesture vocabulary. Classes corresponding to the challenging gestures identified earlier—particularly i, h, r, and i_love_you—show curves that deviate slightly further from the ideal, which is consistent with their lower F1-scores. Conversely, the seven perfect-scoring classes produce ROC curves that approach the top-left corner with minimal deviation, confirming that the model's output probabilities are well-calibrated for these gestures. The overall spread of curves demonstrates that while a small subset of visually ambiguous gestures pulls down the average, the majority of the vocabulary is recognized with high confidence and discriminability. This supports the model's suitability for deployment in scenarios where recognition errors can be managed through confidence thresholding.

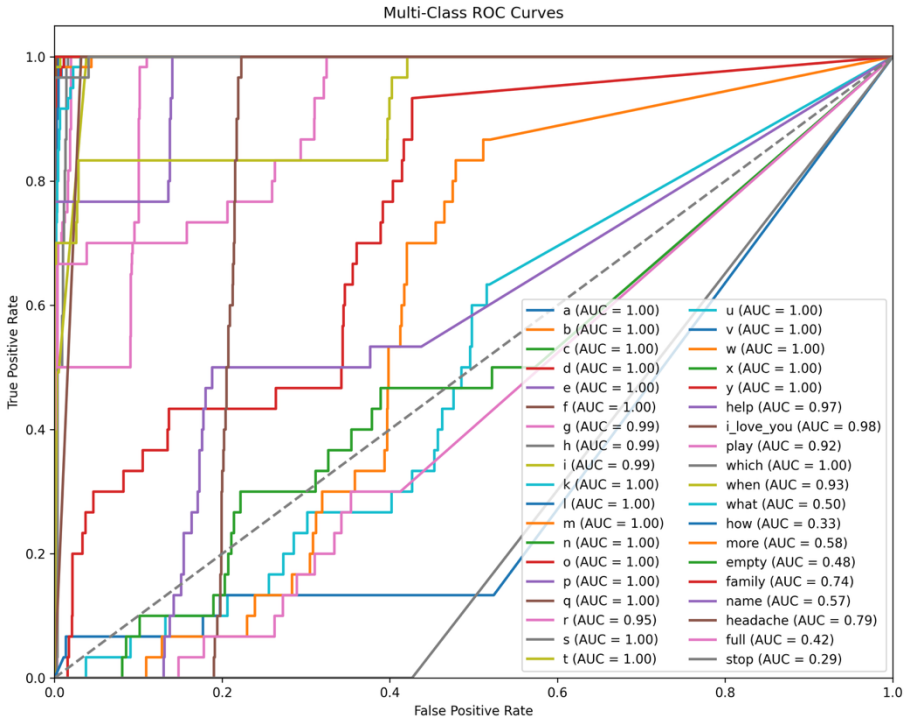


Fig. 8. Multi-class ROC curves. Each curve represents one gesture class; the dashed diagonal indicates random chance.

4.4 Feature Space Visualization (t-SNE)

To gain a deeper understanding of the model's learned feature representations, a t-SNE dimensionality reduction was applied to the 126-dimensional MediaPipe landmark vectors of all test samples (Fig. 9). The resulting 2D projection reveals the structure of the gesture feature space directly from the input data, independent of the classifier. Well-separated, compact clusters correspond to gestures with distinctive hand configurations—such as c, f, p, q, and s—explaining their perfect classification performance. In contrast, the i and i_love_you clusters appear in close proximity with partial overlap in the reduced space, geometrically confirming the source of their persistent confusion. Similarly, the h and r clusters exhibit partial proximity, consistent with their shared two-finger-extended morphology. The t-SNE plot also reveals that conversational gestures such as family and name occupy less compact regions, suggesting higher intra-class variability in their static terminal poses—a plausible explanation for their lower recall values (0.57 and 0.53 respectively). These observations highlight that the primary bottleneck is not the classifier architecture but the inherent geometric ambiguity in the input feature space for a subset of visually similar gestures.

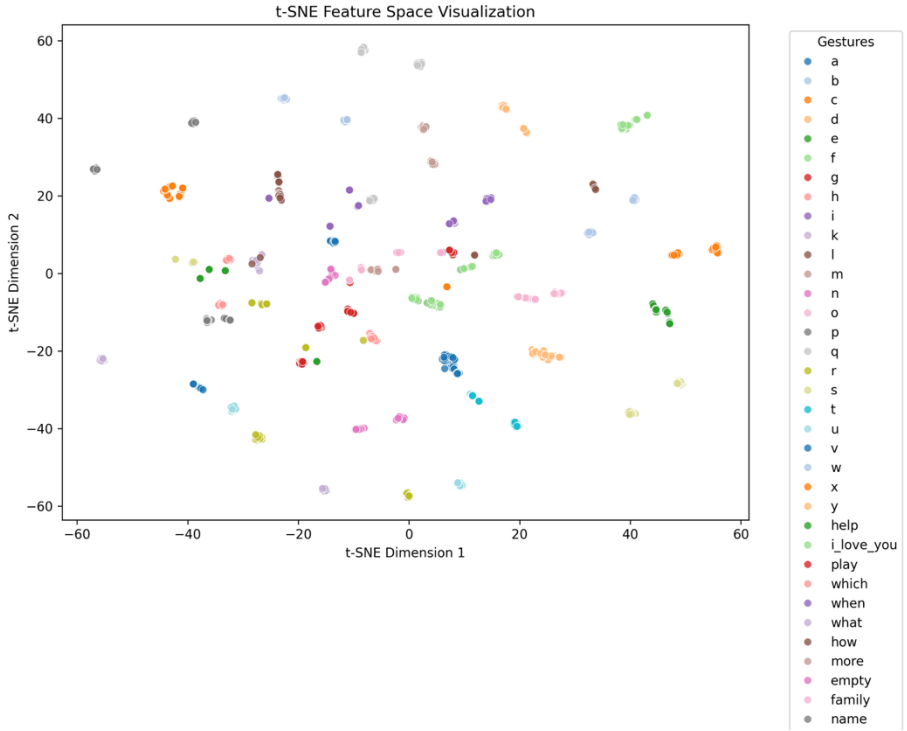


Fig. 9. t-SNE visualization of the 126-dimensional MediaPipe landmark feature space. Each point represents a test sample, colored by gesture class.

4.5 Sensor-Based Glove Evaluation

The smart glove successfully recognized static ASL alphabet letters through flex sensor and IMU data processed on the Arduino Nano and transmitted via Bluetooth to the mobile application. Recognition was functional and produced correct outputs in the majority of cases. However, latency was not entirely consistent across users. Some participants achieved near real-time response (<200 ms), while others experienced slight delays when gestures deviated from the calibrated threshold values. This variability is consistent with the known limitations of rule-based sensor fusion highlighted by Mishra et al. [9] and reflects the challenge of accommodating natural inter-signer variation without adaptive calibration. Crucially, the glove remained robust across different lighting environments, confirming the complementary value of the sensor modality relative to the camera module. Its primary limitation remains the restriction to static gestures, which future LSTM-based integration—as demonstrated in [6, 8]—is expected to address.

4.6 Beta Testing and User Satisfaction

Beta testing with ten participants yielded an overall user satisfaction score of 4.51 out of 5. As summarized in Table 2, Usability and User Experience received the highest rating at 4.80/5, reflecting the system's intuitive navigation and clean interface. General Functionality rated 4.40/5, highlighting reliable operation and a smooth setup process. The Machine Learning component, while positively rated at 4.32/5, showed the most room for improvement, consistent with the model's quantitative limitations for gestures such as i and i_love_you. Feedback from the hard-of-hearing participant specifically highlighted the system's potential utility in daily communication, while noting that occasional false predictions for similar signs could cause slight hesitation. These results collectively confirm that SeGlo delivers a usable and accessible proof-of-concept, and that targeted improvements to challenging gesture classes would most directly enhance user experience.

Table 2. Summarized beta testing ratings

Category	Average Rating (/5)
Usability and User Experience	4.80
General Functionality	4.40
Machine Learning System	4.32
Overall Average	**4.51**

Across all three evaluation categories, ratings remained above 4.30, indicating consistently positive reception of the system. The Usability and User Experience score of 4.80 is particularly noteworthy, as it reflects not only the interface design but also the overall interaction flow between the glove, camera, and mobile application. The relatively lower score for the Machine Learning System aligns directly with the quantitative model findings—specifically the i/i_love_you confusion and the latency variability observed in the glove module—confirming that user perception and empir-

ical metrics point to the same areas for improvement. Importantly, the inclusion of a hard-of-hearing participant provided ecologically valid feedback: the system's text and speech output were described as clear and accessible, while occasional gesture misclassifications were noted as disruptive to communication flow. This feedback reinforces the priority of improving recognition accuracy for visually similar gesture pairs as the single highest-impact step toward practical deployment.

4.7 Comparative Discussion and System Implication

Taken together, these results situate SeGlo within the broader landscape of sign language recognition research. Vision-only systems such as Singh et al. [11] and Alsharif et al. [4] achieve higher raw accuracy (99.12% and 98.2% mAP respectively), but rely on large-scale datasets and computational resources that are incompatible with offline mobile deployment. SeGlo's camera module, achieving 87% overall accuracy on a 38-class vocabulary with a lightweight TensorFlow Lite model, represents a deliberate design trade-off that prioritizes accessibility and portability over peak performance. Glove-only systems validated compatible hardware configurations [6, 7, 9] but lacked mobile integration or remained restricted to rule-based logic. SeGlo advances beyond these baselines by combining both modalities in a unified, offline mobile application. The findings further indicate that the primary failure modes—concentrated in geometrically similar gesture pairs (i/i_love_you, h/r) and dynamic signs evaluated under a static-frame architecture—are well-defined and addressable. Incorporating temporal sequence modeling via LSTM or Transformer-based architectures [3, 8] could resolve the ambiguity of shared static endpoints, while targeted data augmentation for low-performing classes would improve robustness to signer-specific variation. Post-processing with a confidence threshold or rejection class would further reduce false alarms for gestures with low precision. These improvements represent a concrete, evidence-based roadmap for elevating SeGlo from a strong proof-of-concept toward a production-ready assistive communication tool.

5 Conclusion and Recommendation

This study presented SeGlo-12, a dual-mode sign language translation system that integrates a sensor-based smart glove with a vision-based camera module into a single offline-capable mobile application. The system was designed to address the persistent communication gap between deaf and hard-of-hearing individuals and the hearing population by combining the environmental robustness of glove-based sensing with the accessibility and flexibility of computer vision. Evaluation across both modalities confirms that SeGlo achieves this goal as a capable proof-of-concept, while clearly mapping the path toward a production-ready system.

The vision-based module achieved an overall accuracy of 87% and a macro-averaged F1-score of 0.86 across 38 gesture classes on a held-out test set of 1,950 samples, with seven classes reaching perfect classification. Multi-class ROC analysis and t-SNE feature space visualization further validated these findings, revealing that the primary failure modes are concentrated in geometrically similar gesture pairs—

particularly i and i_love_you, and h and r—whose static landmark representations overlap in the input feature space. Conversational gestures with inherently dynamic motion paths, such as family, name, and when, showed lower recall values attributable to the static-frame architecture rather than classifier weakness. The sensor-based glove module demonstrated reliable static gesture recognition and strong environmental robustness, though rule-based threshold logic introduced latency variability across users with differing hand sizes and signing styles. Beta testing with ten participants, including a member of the hard-of-hearing community, yielded an overall satisfaction score of 4.51 out of 5, with Usability and User Experience rated highest at 4.80/5. User feedback corroborated the quantitative findings, identifying occasional misclassifications of similar gestures as the primary friction point in real-world use.

Several concrete directions are recommended for future development. First, replacing the static-frame classification pipeline with a temporal sequence model—such as an LSTM or Transformer architecture applied to landmark sequences—would resolve the fundamental ambiguity among gestures that share a similar endpoint pose but differ in motion trajectory, and would extend the system's vocabulary to dynamic, conversational signs. Second, the glove module should transition from rule-based threshold logic to a machine learning classifier trained on sensor fusion data, following the precedent established by Dalal et al. [6], to improve robustness to inter-signer variation and enable dynamic gesture support. Third, the initially planned hybrid parallel mode—in which glove and camera operate simultaneously for cross-validated recognition—was not realized in this iteration due to time constraints and represents a high-priority next step. Fourth, expanding the training dataset with augmented samples covering varied hand orientations, lighting conditions, and signer demographics would improve generalization, particularly for gestures currently in the moderate-performance tier. Finally, incorporating nonmanual markers such as facial expressions and body pose—as explored by Manasa et al. [8] and recommended by Najib [10]—would bring SeGlo closer to holistic, full-sentence sign language interpretation.

SeGlo's core contribution lies not in achieving maximum classification accuracy in isolation, but in demonstrating that a dual-mode, offline-capable, mobile-first sign language translation system is both technically feasible and positively received by end users, including those from the community it is designed to serve. By prioritizing accessibility, modularity, and real-world deployability, SeGlo advances the field beyond single-modality proof-of-concepts toward an integrated assistive tool with meaningful potential for social impact.

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